Reg. No. | | | | | | | | | | |

[23MSCH1H03]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2024

PHYSICAL CHEMISTRY–I

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. i. Explain partial molar heat content and its significance.

ii. Explain Maxwell relation with respect to Gibbs free energy.

b. i. How does fugacity vary with pressure?

ii. Explain and derive the Clapeyron equation.

c. Explain photoelectric effect. How it supported Plank's concept?

d. Give a graphical representation and expression for rate constants and half life time for first order reaction.

e. What are parallel reactions and how are their kinetics mathematically represented

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. Discuss how quantum mechanics evolved by considering the preliminary discoveries and drawbacks of classical mechanics.

b. What is steady state approximation? Find the rate of any photochemical reaction using steady state approximation.

c. State and explain the factors that influence the transport number. Explain the abnormal transport number.

SECTION – C

3. Answer the following question. **12**

a. i. Explain Nernst heat theorem and its consequences.

ii. Derive fugacity of van der Waal equation.

iii. What is the difference between the kinetic equations of chain, parallel reactions?